

4x8' BETTER THAN 4x4' OR 4x16'

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Knowledgeiswatt

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High intensity training (HIT, that is doing intervals above second threshold/FTP/critical power) is a powerful stimulus to improve endurance performance. However, it requires a great physical and mental effort and leads to fatigue in the following days. As a consequence, it must be carefully dosed and can represent only a small part of the annual training program. So, **it is important understanding how to perform HIT sessions to get the most from this powerful but limited use weapon**

In previous KisW Research Notes, we saw that short (30s on-15s off) high-intensity intervals stimulate superior physiological and performance adaptations compared to long (4-5 minutes) high intensity intervals both in [well-trained](#) and [elite](#) cyclists. When performing the 4-5 minutes long intervals, we saw (read more [here](#)) that a fast start can be beneficial as it permits to ride at higher fraction of VO₂max (a factor which have been linked to HIT efficacy) when compared to keeping a constant pace throughout the 4-5 minutes.

In this KisW Research Note we continue to explore the differences in physiological and performance improvement after different HIT sessions.

Which high intensity long intervals session is better to stimulate physiological adaptations and improve performance? 4x4min, 4x8min or 4x16 min?

A study published by Stephen Seiler and Colleagues (University of Agder, Norway) on *Scandinavian Journal of Medicine and Science in Sports* in 2013 tried to answer this question. (1)

WHAT DID THEY DO?

- They recruited 35 well-trained amateur cyclists (age ~40, VO₂max ~52) which for 7 weeks were divided into 4 different training groups:
 - **Control (CON):** they just rode about 9 hours per week at low intensity.
 - **4x4 Group:** they completed 4hrs at low intensity + 2xweek 4x4 min with 2min recovery HIT sessions.
 - **4x8 Group:** they completed 4hrs at low intensity + 2xweek 4x8 min with 2min recovery HIT sessions.
 - **4x16 Group:** they completed 4hrs at low intensity + 2xweek 4x16 min with 3min recovery HIT sessions.

- All the high intensity intervals sessions were performed at the maximum sustainable intensity for the prescribed duration. These resulted in the following average physiological parameter:
 - 4x16 min: 4.9 mMol/L blood lactate, 100% of second threshold power
 - 4x8 min: 9.6 mMol/L blood lactate, 113% of second threshold power
 - 4x4min: 13.2 mMol/L blood lactate, 131 % of second threshold power

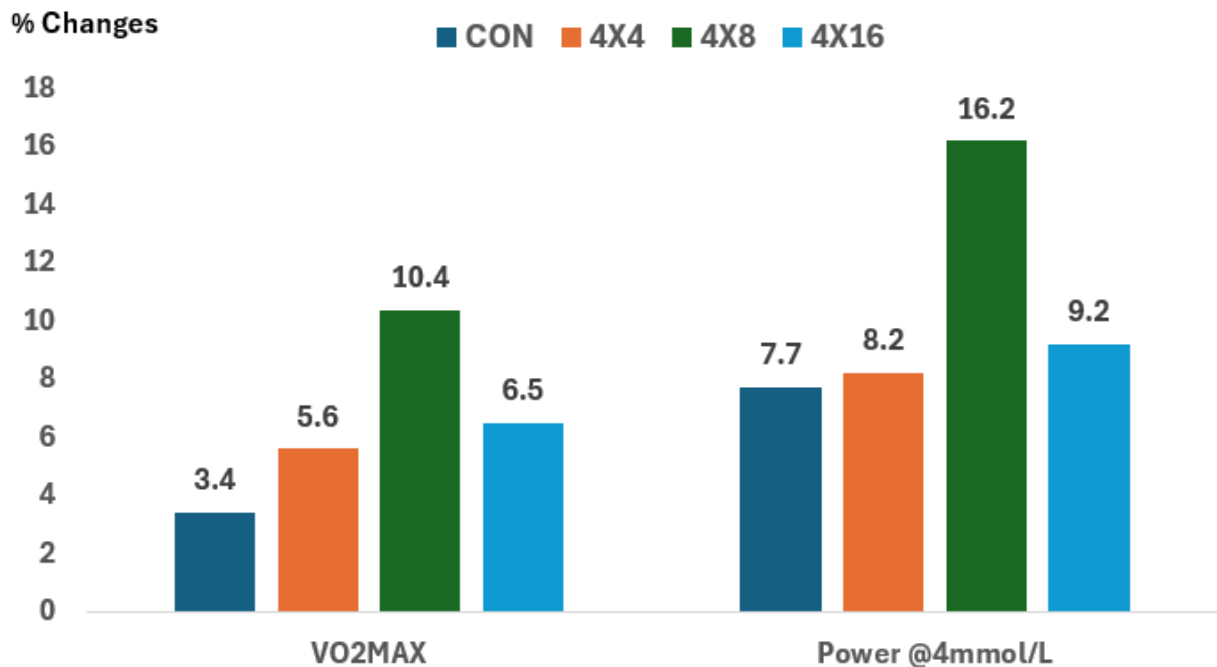
The blood lactate reported is the average of the sample taken after the third and the fourth intervals during all the HIT sessions.

WHAT DID THEY FOUND?

Physiological and Performance changes after the 7 training weeks tended to be superior in 4x8min Group compared to the other groups.

Here the changes on average for each parameter in each group:

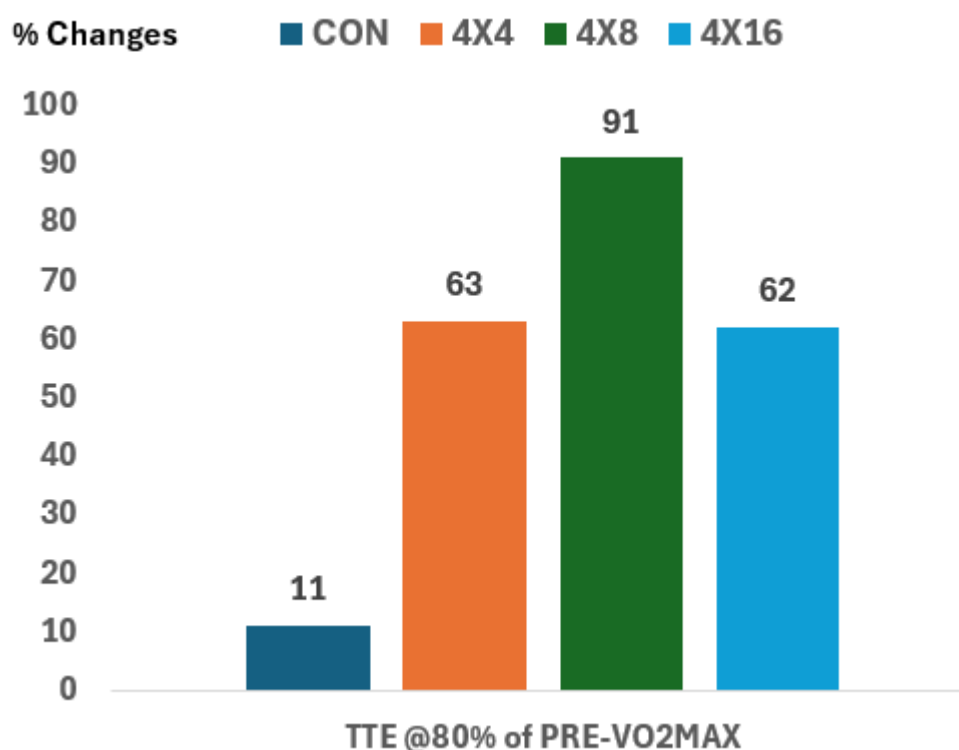
- **VO2max:**
 - CON: from 52.7 to 54.5 ml/min/kg (+3.4%)
 - 4x4: from 50.4 to 53.2 ml/min/kg (+5.5%)
 - **4x8: from 52.8 to 58.3 ml/min/kg (+10.4%)**
 - 4x16: from 51.1 to 54.4 ml/min/kg (+6.5%)
- **Power at 4 mmol/L of blood lactate:**
 - CON: from 222 to 239 W (+7.7%)
 - 4x4: from 220 to 238 W (+8.2%)
 - **4x8: from 241 to 280 W (+16.2%)**
 - 4x16: from 228 to 249 W (+9.2%)



% changes in Vo2max and power output at 4 mmol/L of blood lactate concentration in the four different groups. Data from Seiler et al. (1).

Time to Exhaustion at 80% of VO2max power (VO2max power was calculated before starting the 7 training weeks):

- CON: from 10.9 to 12.1 min (+11%)
- 4x4: from 9.7 to 15.8 min (+63%)
- **4x8: from 11.9 to 22.7 min (+91%)**
- 4x16: from 8.5 to 13.8 min (+62%)



% changes in time to exhaustion at 80% of VO2max calculated before starting the 7 weeks. Data from Seiler et al. (1)

PRACTICAL TAKEAWAYS:

When considering long high intensity intervals performed at the maximum sustainable intensity, 4x8 min seems a better solution compared to 4x4min and 4x16min to induce superior aerobic physiological and performance adaptations.

GABRIELE'S COMMENT

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- **These data show another effective weapon to add to our high intensity arsenal: 8min intervals.** While in this study they were performed at the maximum sustainable intensity (which resulted in 113% of second threshold), I would suggest to lower a little bit the intensity in order to be able to repeat the workout many times even when the weekly training load is high. I would say that 105-110% can be enough to stimulate adaptations, but this is just a personal consideration.

- When implementing this intervals during a training plan, I would consider the following basic training principles:
 - **progressive overload:** starting from a 2x8 and then progressing to 3x8 and 4x8 can be a good strategy (I would not exceed 4 intervals during the same workout).
 - **specificity:** I see this workout more specific (and so more effective to apply) for road cycling climbers/time trialists or for mountain bike marathon, as it mimics more the long steady pace efforts of these races. For cyclocross and MTB XCO the 30-15 short intervals saw previously could mimic more the specific race demands and so they can be more effective.

Thanks for reading! Share with your team!

Gabriele Gallo, PhD in Exercise and Sport Sciences

REFERENCES

1. Seiler S, Jøranson K, Olesen BV, Hetlelid KJ. Adaptations to aerobic interval training: interactive effects of exercise intensity and total work duration. Scand J Med Sci Sports. 2013 Feb;23(1):74-83.



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